

Nonlinear Difference-in-Differences with Repeated Cross Sections[†]

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I extend to repeated cross sections the nonlinear difference-in-differences (DID) methods in Wooldridge (2023) for staggered interventions with panel data. The key is a conditional parallel trends (CPT) assumption imposed on a strictly increasing function of the conditional mean. As with panel data, one can apply pooled quasi-maximum likelihood estimation (QMLE) to log likelihoods in the linear exponential family (LEF) with cohort and time dummies included flexibly. The resulting estimators are fully robust to distributional misspecification, only requiring correct specification of the conditional mean.

I show how to obtain both lags only (LO) estimators, where only current and lagged treatment indicators are included, and leads and lags (L&L) estimators within one framework.

The setting here extends the linear analysis in Deb et al. (2025)—henceforth, DNWZ. Callaway and Sant’Anna (2021)—henceforth, CS—propose doubly robust estimators that combine regression and propensity score weighting, but the CPT assumption is stated in levels. The same is true in Borusyak, Jaravel, and Spiess (2024) (who propose a two-step imputation estimator) and DNWZ (who show that imputation and pooled ordinary least squares are identical under certain conditions).

I. Assumptions, Definitions, and Identification

The population—say, individuals or firms—is assumed to consist of units that exist over T time periods, indexed by $t = 1, 2, \dots, T$. We assume that the population is stable, but the distributions of variables are allowed to change across t .

An intervention occurs for the first time in period q with $1 < q \leq T$, with additional units subjected to the intervention in subsequent periods. As in Wooldridge (2023), we define potential outcomes by treatment cohort,

$$(1) \quad Y_t(g), g \in \{q, \dots, T, \infty\}, t \in \{1, 2, \dots, T\},$$

where g indicates the first time subjected to the intervention (the treatment cohort) and t is calendar time. The case $g = \infty$ indicates the potential outcome in the never-treated state.

We let $\{D_g : g = q, \dots, T, \infty\}$ be exhaustive and mutually exclusive cohort indicators expressing the first period a unit is subjected to the intervention; these do not vary across time. The dummy D_∞ denotes the never-treated (NT) group. Here we assume that there is an NT group; this restriction is easy to relax along the lines discussed in Wooldridge (2023).

The initial goal is to estimate average treatment effects on the treated (ATTs) separately by cohort, g , and treated periods $t = g, \dots, T$:

$$(2) \quad \tau_{gt} = E[Y_t(g) - Y_t(\infty) | D_g = 1] = E[Y_t | D_g = 1] - E[Y_t(\infty) | D_g = 1],$$

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where the equality follows from $Y_t = Y_t(g)$ when $D_g = 1$. These parameters are the main focus in the DID literature that assumes linear outcome models.

A set of observed covariates can be indexed by time: $\{\mathbf{X}_t = 1, \dots, T\}$, where $\mathbf{X}_t$ is a $1 \times K$ vector. The $\mathbf{X}_t$ are not indexed by potential outcomes, $\mathbf{X}_t(g)$, because we assume that the covariates do not change with the treatment assignment; that is, we impose a no bad controls assumption on the covariates. Often the underlying covariates will not vary over time, but we cannot know for sure because we observe each unit for at most one time period. We assume no spillover effects in that only a unit's assigned treatment status affects its outcome. This stable unit treatment value assumption (SUTVA) is often plausible in studies relying on repeated cross sections because the policy assignment is often at a higher level than the unit, such as the census tract or county. See DNWZ for additional discussion.

The following assumptions are adapted from Wooldridge (2023, 2025).

ASSUMPTION CNA (Conditional No Anticipation): For $g \in \{q, \dots, T\}$ and $t \in \{1, \dots, g-1\}$,

$$(3) \quad E[Y_t(g) | \mathbf{D}, \mathbf{X}_t] = E[Y_t(\infty) | \mathbf{D}, \mathbf{X}_t].$$

A sufficient condition is $Y_t(g) = Y_t(\infty)$, $g \in \{q, \dots, T\}$, $t \in \{1, \dots, g-1\}$, which can be violated if units that are eventually part of cohort g anticipate the intervention and change their behavior.

ASSUMPTION CPT (Conditional Parallel Trends): For $t = 1, 2, \dots, T$ and $G(\cdot)$ a twice-continuously differentiable, strictly increasing function,

$$(4) \quad E[Y_t(\infty) | \mathbf{D}, \mathbf{X}_t] = G\left(\alpha + \sum_{g=q}^T \beta_g D_g + \mathbf{X}_t \boldsymbol{\kappa} + \sum_{g=q}^T (D_g \cdot \dot{\mathbf{X}}_{tg}) \boldsymbol{\eta}_g + \gamma_t + \mathbf{X}_t \boldsymbol{\pi}_t\right),$$

where $\dot{\mathbf{X}}_{tg} \equiv \mathbf{X}_t - E(\mathbf{X}_t | D_g = 1)$ are the covariates centered about cohort/time period means and $\mathbf{D} = (D_q, \dots, D_T)$ is the vector of treatment indicators. As normalizations, $\gamma_1 = 0$, $\boldsymbol{\pi}_1 = \mathbf{0}$.

As discussed in Wooldridge (2023), Assumption CPT imposes the parallel trends assumption on $G^{-1}(E[Y_t(\infty) | \mathbf{D}, \mathbf{X}_t])$. In general, CPT does not hold for $E[Y_t(\infty) | \mathbf{D}, \mathbf{X}_t]$. There are two exceptions. First, if there is no selection into treatment in the sense that $\beta_g = 0$, $\boldsymbol{\eta}_g = \mathbf{0}$, $g = q, \dots, T$, then $E[Y_t(\infty) | \mathbf{D}, \mathbf{X}_t] - E[Y_1(\infty) | \mathbf{D}, \mathbf{X}_1]$ does not depend on $\mathbf{D}$. In this case, cohort assignment is unconfounded with respect to the level $Y_t(\infty)$ conditional on $\mathbf{X}_t$. A second case where CPT holds for $E[Y_t(\infty) | \mathbf{D}, \mathbf{X}_t]$ is when the covariates do not change over time and the coefficients γ_t and $\boldsymbol{\pi}_t$ do not vary over time—a stationarity restriction.

The reason for centering the covariates in the interaction $D_g \cdot \dot{\mathbf{X}}_{tg}$ is to make the β_g interpretable as average selection effects. This is most easily seen in the linear case, in which case β_g measures the average difference in the mean outcome of the $Y_t(\infty)$ for cohort g and the NT cohort. Generally, β_g measures this difference based on other transformations, such as the natural logarithm if G is exponential. This centering does not affect estimation of the ATTs.

The terms $\mathbf{X}_t \boldsymbol{\pi}_t$ allow relaxation of the usual parallel trends assumption even if the $\mathbf{X}_t$ do not change over time. Their inclusion plays the same role as in CS, who apply standard treatment effects estimators to a linear outcome equation when covariates are available.

The observed outcome in period t can be written

$$Y_t = D_q Y_t(q) + D_{q+1} Y_t(q+1) + \dots + D_T Y_t(T) + D_\infty Y_t(\infty).$$

In the staggered intervention case without exit, the time-varying treatment indicator is

$$W_t = D_q \cdot p q_t + D_{q+1} \cdot p(q+1)_t + \dots + D_T \cdot p T_t,$$

where $p g_t = f g_t + \dots + f T_t$ and the $f s_t$ are time period dummies.

Similar to Wooldridge (2023), under Assumptions CNA and CPT, the parameters in (4) are identified using the untreated observations. Specifically, in the subpopulation of untreated units at time t , we can write

$$(5) \quad E(Y_t | \mathbf{D}, \mathbf{X}_t, W_t = 0) = G \left(\alpha + \sum_{g=q}^T \beta_g D_g + \mathbf{X}_t \boldsymbol{\kappa} + \sum_{g=q}^T (D_g \cdot \dot{\mathbf{X}}_{tg}) \boldsymbol{\eta}_g + \sum_{s=2}^T \gamma_s f s_t + \sum_{s=2}^T (f s_t \cdot \mathbf{X}_t) \boldsymbol{\pi}_s \right),$$

where we use Assumption CNA along with Assumption CPT. The no bad controls assumption mentioned above and SUTVA are also implicit in (5).

II. Estimation by Pooled QMLE

Equations (2) and (5) immediately suggest an imputation approach to estimation, as in Wooldridge (2023, Procedure 1), where the parameters in (5) are estimated using only the control observations. A downside to imputation is that obtaining standard errors is complicated by the two-step estimation. Instead, a pooled approach is attractive. Assume the availability of random samples from the population at each time period t . The draws, indicated by i , are independent, but not generally identically distributed (because the population distribution of the $Y_t(g)$ and $\mathbf{X}_t$ may change across t). For unit i , $t(i)$ represents the observed time period. The observed data for unit i are $Y_{i,t(i)}$, $\mathbf{X}_{i,t(i)}$, and the cohort indicators $D_{i,t(i),g}$.

As discussed in Wooldridge (2023) in the panel data case, certain combinations of LEF log likelihoods and conditional mean functions render imputation identical to pooled QMLE; the same is true with repeated cross sections. The canonical link pairs include, among others, (i) the linear mean function and normal log likelihood, (ii) the logistic mean function and Bernoulli log likelihood, and (iii) the exponential mean function and Poisson log likelihood.

The pooled QMLE uses a conditional mean function for all draws across all time periods:

$$(6) \quad E[Y_{i,t(i)} | \mathbf{D}_{i,t(i)}, \mathbf{X}_{i,t(i)}] = G \left[\alpha + \sum_{g=q}^T \beta_g D_{i,t(i),g} + \mathbf{X}_{i,t(i)} \boldsymbol{\kappa} + \sum_{g=q}^T (D_{i,t(i),g} \cdot \dot{\mathbf{X}}_{i,t(i),g}) \boldsymbol{\eta}_g + \sum_{s=2}^T \gamma_s f s_t + \sum_{s=2}^T (f s_t \cdot \mathbf{X}_{i,t(i)}) \boldsymbol{\pi}_s + \sum_{g=q}^T \sum_{s=g}^T \delta_{gs} (W_{i,t(i)} \cdot D_{i,t(i),g} \cdot f s_t) + \sum_{g=q}^T \sum_{s=g}^T (W_{i,t(i)} \cdot D_{i,t(i),g} \cdot f s_t \cdot \dot{\mathbf{X}}_{i,t(i),g}) \boldsymbol{\xi}_{gs} \right].$$

The first line in this equation accounts for possible selection effects into treatment, allowing $Y_{i,t(i)}(\infty)$ to be systematically correlated with cohort $\mathbf{D}_{i,t(i)}$. The second line accounts for secular trends in $Y_{i,t(i)}(\infty)$ along with heterogeneous trends $f s_t \cdot \mathbf{X}_{i,t(i)}$. The final line includes the treatment indicators entered in a flexible way, with moderating effects $\boldsymbol{\xi}_{gs}$ in the linear index.

PROCEDURE 1:

- (i) Using all of the data, use pooled QMLE in the LEF with the canonical link function to estimate the parameters in (6).
- (ii) Obtain the $\hat{\tau}_{gt}$ as the average partial effects (APEs) with respect to $W_{i,t(i)}$ averaged over the pairs (g, t) , being sure to recognize $W_{i,t(i)}$ as a binary variable.

The estimates $\hat{\delta}_{gs}$, while not the $\hat{\tau}_{gs}$ except in the linear case, are often of interest. Centering the covariates in the last line of (6)—that is, using $\dot{\mathbf{X}}_{i,t(i),g}$ in the interactions—is important to give meaning to the $\hat{\delta}_{gs}$. When $G(\cdot)$ is the logistic function, the $\hat{\delta}_{gs}$ are ATTs on the log odds; when $G(\cdot)$ is the

exponential function, the $\hat{\delta}_{gs}$ are proportionate ATTs. In order to compare quantities across different models, we need to produce the $\hat{\tau}_{gs}$.

III. L&L (Event Study) Estimation

Procedure 1 is an LO method because it only includes dummy variables representing treatment periods. As a way to test the CPT assumption, we can add pretreatment indicators along with their interactions with the covariates. In (6), this means adding, for each treated cohort g ,

$$(7) \quad D_{i,t(i),g} \cdot fs_{t(i)} \text{ and } D_{i,t(i),g} \cdot fs_{t(i)} \cdot \mathbf{X}_{i,t(i),g}, s = 1, \dots, g-2,$$

where it is traditional to drop the terms involving $D_{i,t(i),g} \cdot fs_{t(i)}$ for $s = g-1$, which sets $g-1$ as the reference period. In the linear case, this expanded regression produces a version of the L&L estimator in DNWZ.

A test for pre-trends typically centers on the significance of the variables in (7) not involving $\mathbf{X}_{i,t(i),g}$. With the covariates centered, the coefficients on the $D_{i,t(i),g} \cdot fs_{t(i)}$ are average pre-trend estimates by cohort/time.

Generally, we cannot rank the efficiency of the LO and L&L estimators, although the LO often has an advantage because it uses fewer parameters. Also, one cannot claim that one estimator has less bias than the other when CPT is violated. In addition to comparing a linear model to a nonlinear model that exploits the nature of Y_t , it is a good idea to compare LO and L&L versions, as these have different sensitivities to violation of CPT.

IV. Practical Considerations and Extensions

A. Clustering Standard Errors and Inference

Above, we assumed that the observations are independent across (i) of Procedure 1. In this scenario, one only needs to make inference robust to misspecification of the assumed LEF density. In the linear case, this leads to the usual heteroskedasticity-robust standard errors.

Often a case can be made to cluster standard errors with repeated cross-sectional data. For one, the data may be obtained from a cluster sampling scheme. For example, if census tracts are sampled and then people are sampled from the tracts, one should cluster at the census tract level. If stratified sampling is used, it is often straightforward to apply the pooled estimation method in Procedure 1 with sampling weights and clustering.

Even without a cluster sampling scheme, one may need to cluster standard errors to account for the treatment assignment—a case studied in Abadie et al. (2023). For example, a treatment might be assigned at the census tract level, where all units are assigned the same cohort D_g . In order to reflect the uncertainty due to clustered assignment, the standard errors should be clustered at the census tract level.

B. Aggregating the Treatment Effects

Rather than report a full set of ATT estimates for each treatment cohort, it is common to aggregate the effects by exposure time and, in the case of L&L, time until exposure. In the balanced panel data case, one typically weights the estimates $\hat{\tau}_{gt}$ by the relative cohort shares, which vary by exposure time in the staggered case. One can also obtain a single weighted average of all of the $\hat{\tau}_{gt}$ across treated periods; see Wooldridge (2025). In the L&L cases, we can also weight the leads by cohort shares to produce a single event study plot, where $\hat{\tau}_{g,g-1} = 0$ as the normalization.

With repeated cross sections, the weighting issue is more subtle, as discussed in CS. One simple strategy is to compute weights that vary by t as well as by g , based on sample sizes N_{gt} , the number of units in cohort g in period t . Choosing weights in this way is equivalent to computing the APE with respect to $W_{i,t(i)}$ and then aggregating by the appropriate subsample. For example, to obtain the

estimates by exposure time in the LO case, one computes the APE with respect to $W_{i,t(i)}$ in the pooled estimation and then averages over all observations corresponding to exposure times of zero, one, two, and so on. In the L&L case, one can get the aggregated pre-trend effects by introducing a new binary indicator, $NW_{i,t(i)} = 1 - W_{i,t(i)}$, and multiplying it by the pretreatment terms in (7). Then, compute the APE with respect to $NW_{i,t(i)}$ and average over all observations corresponding to times until treatment.

In the nonlinear case, there is a subtle point about obtaining an aggregate version of the event study graph. Namely, the CPT assumption holds on the linear index, not necessarily the conditional mean. To detect violations of CPT, it makes sense to aggregate the $\hat{\delta}_{gs}$ in an event study plot. In leading cases, this will produce aggregated effects on the log odds (logit) or on the log of the mean (exponential).

Statistical packages, such as Stata, that have features for computing APEs also will report standard errors of the weighted effects. Even if the standard errors are not clustered, we still need to know that there are sufficient observations within each cohort/time period cell in order for the standard errors to be reliable. In some cases, we may need to impose restrictions directly on the model rather than estimating the most flexible LO or L&L model. For example, if some treated cohorts have only a small number of units, we might impose constant effects by exposure time, which means collapsing the dummies $D_g \cdot fs_t$ into dummies that vary only by $s - g$, $s = g, \dots, T$ (LO) and then adding $s = 1, \dots, g - 2$ for L&L. In extreme cases, we might need to impose a constant effect, which means replacing all treatment dummies with the single indicator $W_{i,t(i)}$. We can always compare this coefficient with the weighted average from the fully flexible estimation to see if imposing a constant effect changes the results. The reason for imposing a constant effect is so that we can better trust the inference.

C. Heterogeneous Time Trends

One approach to accounting for violations of PT that occur even after including covariates is to assume relative simple heterogeneous trends in the absence of the intervention. In particular, with at least two pre-intervention periods per treated cohort, one can include in (6) $D_{i,t(i),g} \cdot t$, and even $D_{i,t(i),g} \cdot t \cdot \mathbf{X}_{i,t(i),g}$. The coefficients on the $D_{i,t(i),g} \cdot t$ can be used to test for pre-trends. Like in the panel data case in Wooldridge (2023), such tests do not suffer from contamination bias, provided that the covariates are included flexibly, as in (6).

Including cohort-specific trends can be costly in terms of precision because their inclusion creates collinearity with the treatment indicators. Of course, using a pretest to decide whether to drop these terms can be problematic—just as when using the L&L approach.

D. Empirical Application

Due to space constraints, I do not include an empirical illustration. An analysis using the data in DNWZ, which includes more than 500,000 individual observations with assignment at the Public Use Microdata Areas level (of which there are 682), is available on request from the author—including Stata code. The intervention is adoption of the Secure Communities immigration enforcement program in the United States, and the outcome variable is participation in the Supplemental Nutrition Assistance Program, a government nutritional program. The logit model proposed here is compared with a linear probability model similar to that used in DNWZ.

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